

Radio Cartographer Tutorial

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The Basics

- The Task Bar (for reference):

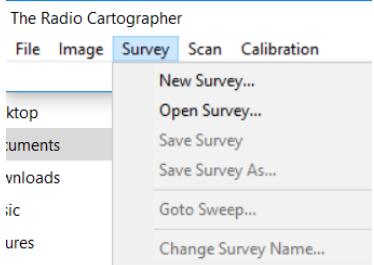
The Radio Cartographer

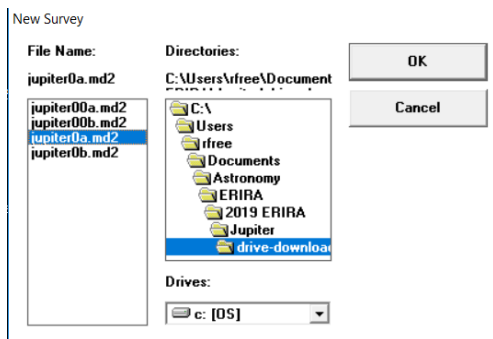
File Image Survey Scan Calibration

- Make sure you don't use any channel b data or calibration files (example B channel file name: jupiter00**b**.md2).
- This tutorial is specific for **Surveys**. Surveys utilize .md2 files. **Scans** use .md1 files. If you click "New Survey" and you're unable to locate a data file that you know you've downloaded, it's probably because the file you downloaded is a .md1 file, in which case you need to open the file by clicking "Scan → New Scan."

New Survey

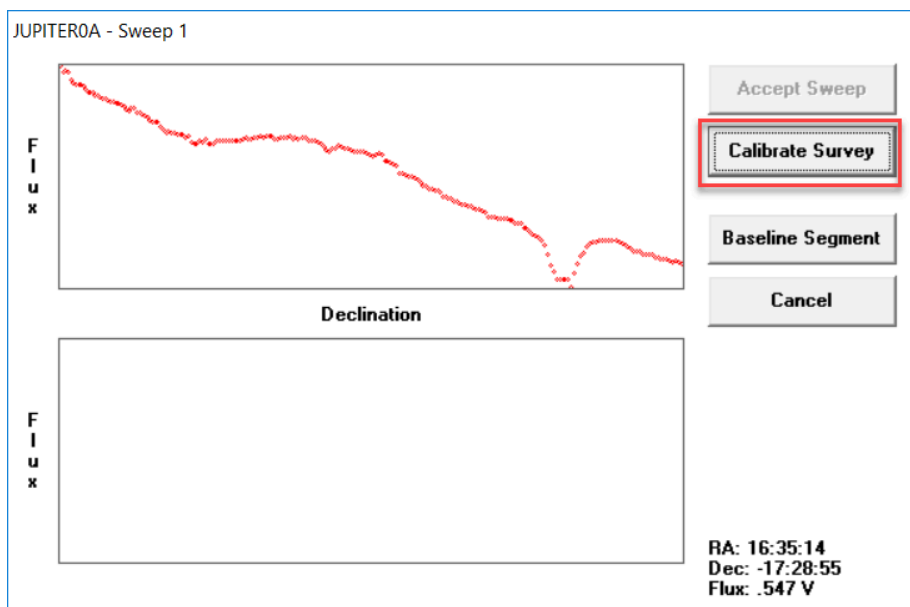
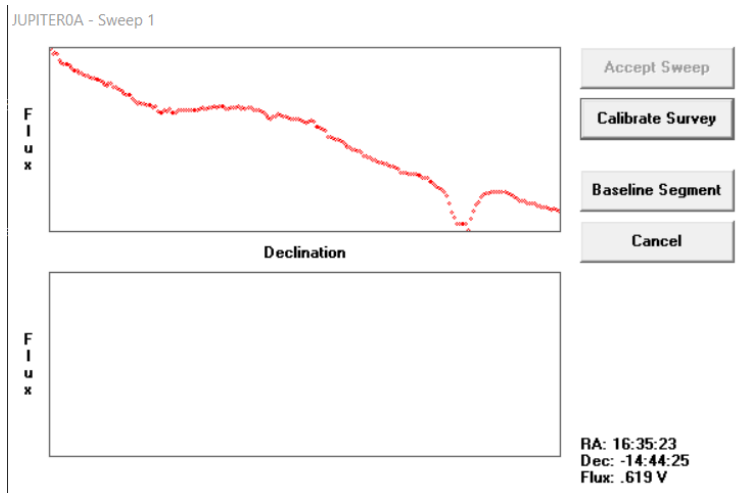
1. New survey → select data file you would like to work with





Calibrating

2. Calibration → select calibration → select cal18a.cal → calibrate survey

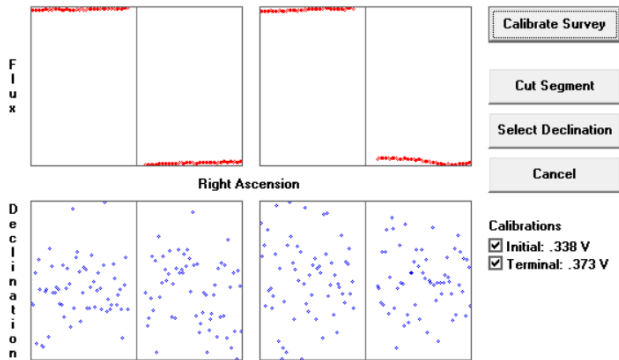


Cutting Segments

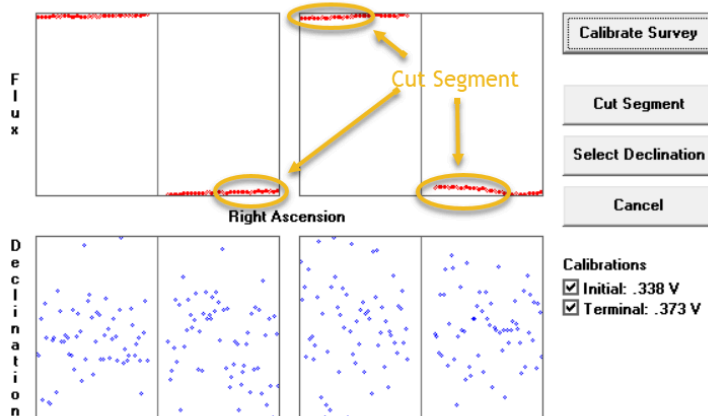
3. Cut any segments that contain a “jump” or a “drop.” Additionally, cut excess data (see orange circles). Do cut segments: click “cut segment” → drag mouse from cutting point to edge of plot → double click. [Note: the part that’s highlighted green will be the part that is cut out.]

4. When you’re done cutting your segments, click “calibrate survey.”

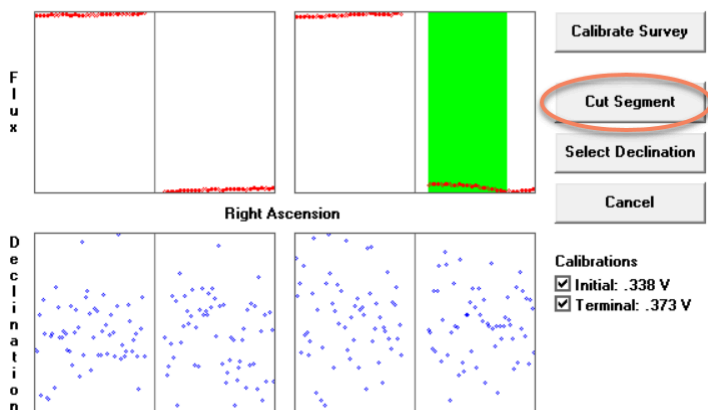
JUPITER0A - Sweep 1



JUPITER0A - Sweep 1



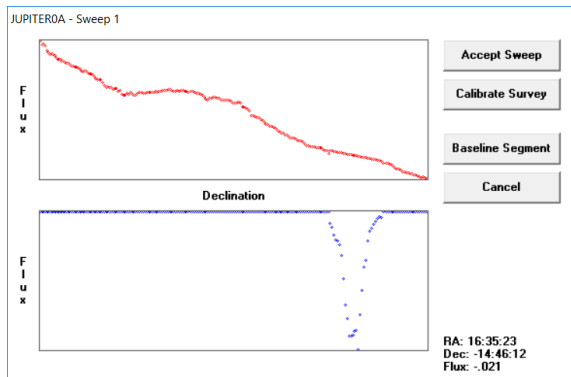
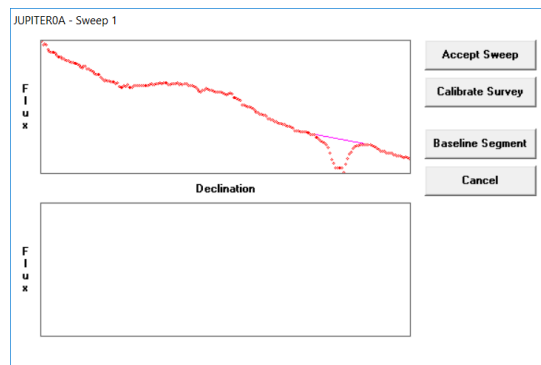
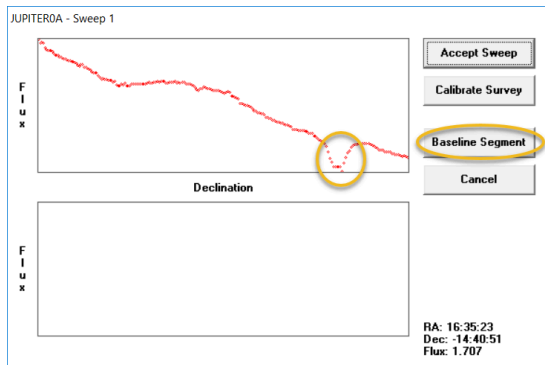
JUPITER0A - Sweep 1



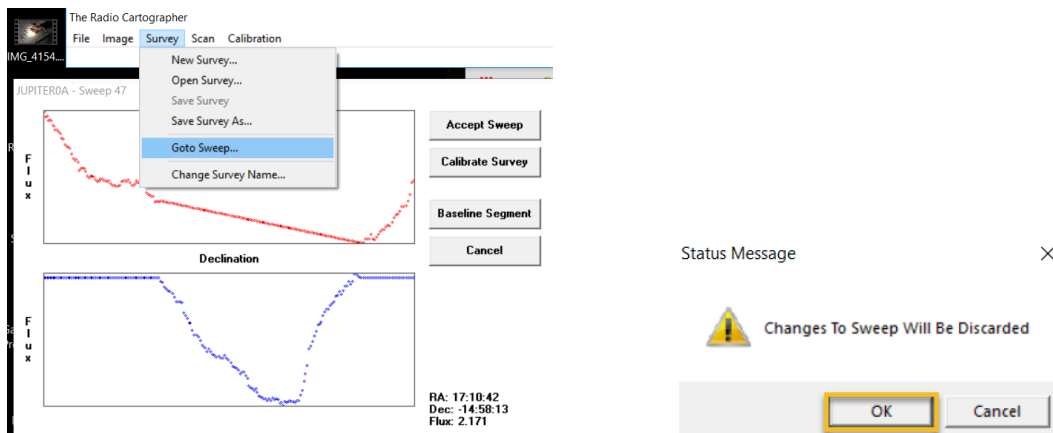
Accepting Sweeps

Next we will go through each sweep individually. You will want to cut out any “spikes” in the data (these are due to radio interference, RFI, and will distort our data), or any “drop outs.” Your sweeps should look relatively smooth, unless there is a known source causing a spike.

5. To cut out RFI areas, click “baseline segment” → drag the mouse from the beginning of the area you want to cut to the end of the area → click. Segment should be cut from the sweep. Do this for each sweep.



- To undo a cut, go to “survey” → “GoTo Sweep” → click “ok” → click “ok.” You can also use this to go back to a sweep if you need to.



- To cancel a cut in progress, simply right click your mouse.

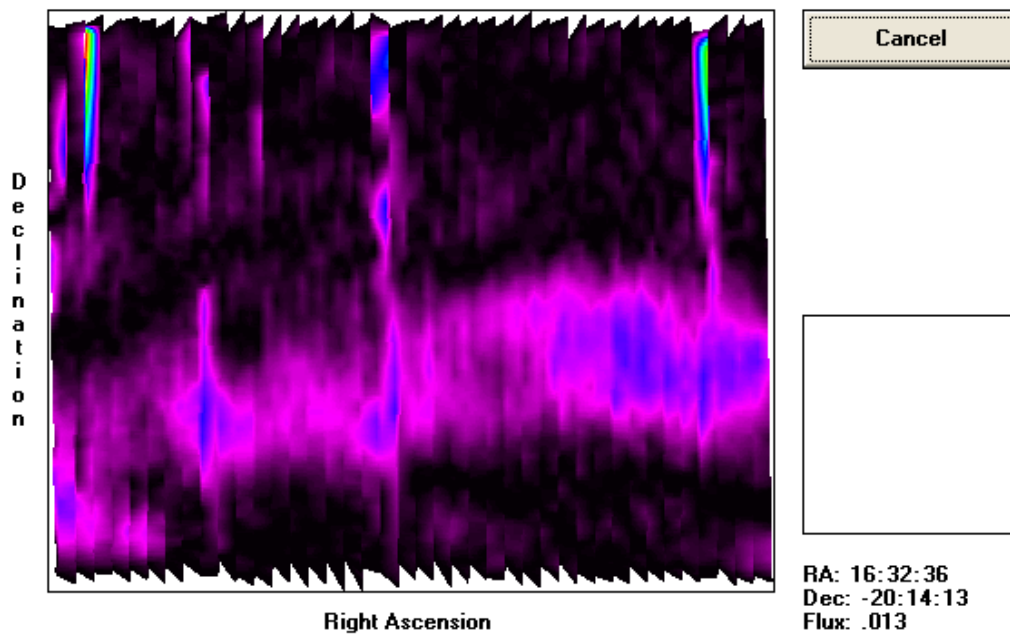
If you come across a sweep that is lacking data points, just move on.

Once you've gone through all of the sweeps, the program will prompt you to "Save Changes to Open Survey." Click yes, and save the file to your computer.

The Image

Next we will work on making the actual image. Do not click the "make image" button until the very end of this section.

6. We'll begin by smoothing the data. To do this, simply click "smooth sweeps" → "ok."
7. Now we'll play with the baseline sweeps. Click "baseline sweeps" → click "ok."
8. Next we'll align the sweeps. Click "align sweeps" → change the prompted number (3) to "0.5" → click "ok".
9. Click "make image" → change the prompted number (2) to "1" → click "ok."

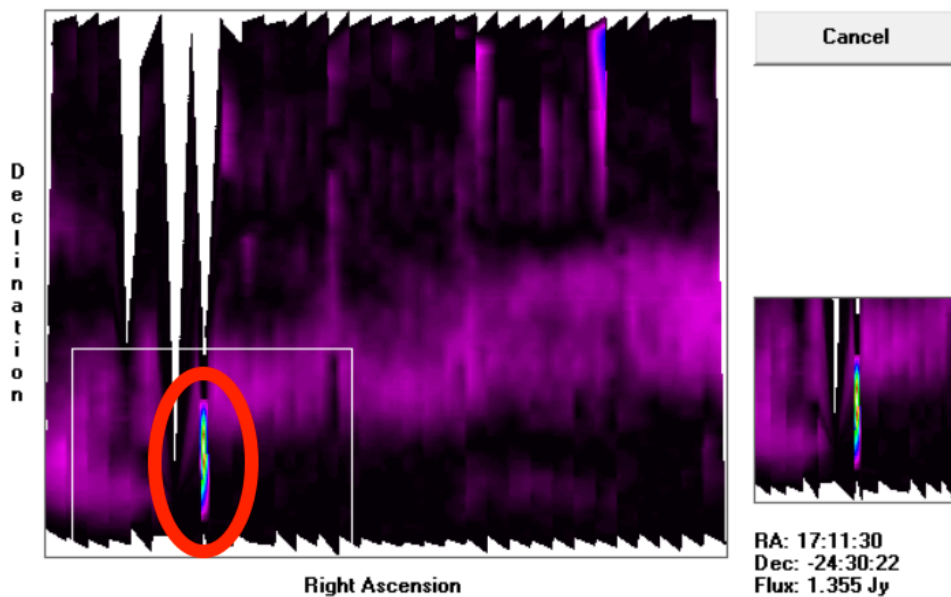


Note: if you scroll your mouse over the image, you can attain data about the RA, Dec, and Flux by looking at the bottom right hand corner of the screen.

The Palettes

More often than not, the images will still have portions of the data skewed by RFI. We can use the palettes to take care of this issue (example below):

JUPITER0A - Image



It's also important to note that we are not longer examining a survey, we are examining an image. So when you're working with the task bar, you'll want to use the options under "image," not the ones under "survey."

10. First we need to determine what the maximum value of the true data (colored purple) is. The easiest way to do this is to hover your mouse over the “most purple” portion of the image and recording the value for flux in Janskys at this location.

- NOTE: In your image, the maximum might fall in a color other than purple. Take the maximum of your specific source.

11. Click “image” → “show palette.” Replace the “max” value of the Flux Range (in the upper left corner) with a value slightly higher than the true maximum from step 10. Click “ok.”

- NOTE: You will have difficulty typing in the number you want. It happens. There’s no way around it. Struggle through it!

12. To save your final image, click “image” → “save image as...”

Congratulations! You’ve finished creating and refining your image with Radio Cartographer!

Convert flux to Janskys

“Calibration” → “Select Calibration” → open recent calibration file

